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Table of Contents

1. Introduction	3
2. Overview of major hardware components	3
3. company profiles and technologies	4
3.1 NVIDIA	4
3.2 Dell Technologies	7
3.3 Micron Technology	8
3.4 Toshiba	10
4. Integration in a computer system	12
5.Future trends	12
6.Recommendations	13
7. Conclusion	13
8. References	14

1. INTRODUCTION

Modern computer architecture is based on a set of simple components, such as computer processors, memory (especially random-access memory) , storage, and computer systems and each of them performs a basic function. They execute tasks, provide data processing capabilities and allow computer systems to perform a useful task.

The purpose of this document is to help us understand the main hardware components of modern computer systems and the role of major technology companies in the hardware market. This document focused on **NVIDIA, Dell Technologies, Micron Technology, and Toshiba** that manufacture processors, computer systems, memory, and storage technologies, respectively.

This document will provide a description of what these components are, how they are used, and how they coordinate with each other in modern workloads.

2. OVERVIEW OF MAJOR HARDWARE COMPONENTS

➤ Processors (CPU/GPU)

The processor carries out instruction processing and arithmetic calculations.

The Central Processing Unit (CPU) is the main processor, which is effective for general-purpose applications.

The Graphics Processing Unit (GPU) is a specialized processor for parallel processing, and has been widely successful for graphics rendering, as well as applications such as artificial intelligence. Companies such as NVIDIA have developed GPUs for high-performance computing applications.

➤ Computer systems and manufacturers

A computer system combines a computer such as a desktop computer, laptop computer or server with all its hardware components into one system.

Computer systems are products that manufacturers are expected to design and build to meet a range of user and business needs. Dell Technologies is one of the leading suppliers of these systems and has a large portfolio.

➤ **Memory (RAM)**

Main memory (or random-access memory, RAM) holds the data and instruction sets used by a computer processor. This helps speed the performance of a computer as it can quickly access the data and run multiple applications simultaneously.

Micron Technology is one of the companies that builds memory products to improve the speed of computing.

➤ **Storage Devices**

Storage devices are components that permanently store applications, documents, operating systems, and other data.

Common types of storage devices include Hard Disk Drives (HDDs) and Solid-State Drives (SSDs). They are intended to protect data from being lost when the power is turned off.

Toshiba has developed many technologies to provide stable high-capacity storage.

3. COMPANY PROFILES AND TECHNOLOGIES

➤ **NVIDIA**

A. Background and History

NVIDIA is a US-based technology which was founded on April 15, 1993 by Jensen Huang, Chris Malachowsky and Curtis Priem in California and it has been widely described as a Big Tech company.

It is an American technology company best known for inventing the modern graphics processing unit (GPU) and becoming the dominant hardware platform for artificial intelligence and accelerated computing.

The company initially focused on developing graphics processing technology for video games. In its early stage, many difficult situational barriers were encountered by Nvidia, which included product failing and monetary struggles. Even though it struggled, the launching of the RIVA 128 and GeForce units assisted the company to reach high levels of success rapidly.

As time progressed, Nvidia expanded gaming into artificial intelligence (AI), high-performance computing, and also massive digital data storage centers.

It has been observed that Nvidia stands as a primary international technical group for GPU and AI innovation inside this era.

B. Key Products and Technologies

Nvidia is known for lots of advanced hardware and software technologies and it creates many different kinds of high-tech hardware and digital code for computers today.

- Graphics Processing Units (GPUs):

Different products like GeForce which used for gaming purpose and RTX tools are widely used for graphics rendering and computing.

- CUDA Platform:

The CUDA system helps Nvidia processors (GPU) to do many tasks at the very same moment and enables them to perform parallel processing in order to make them powerful for AI and scientific computing.

- Tegra Processors:

Tegra is used in mobile devices and automotive systems.

- AI and Data Center Solutions:

Products like DGX systems and specialized electronic components (AI chips) are used for deep learning, machine learning, and manage huge amounts of information.

- Autonomous Driving Technology:

The Nvidia Drive platform supports self-driving cars with the help of AI.

All these inventions have helped Nvidia to become a top company in both hardware and software innovation.

C. Roles in Modern Computing

Nvidia plays a critical role in modern computing systems:

- Artificial Intelligence (AI):

Because of their high-speed parallel processing capabilities, Nvidia GPUs are widely used to train and deploy AI models.

- High-Performance Computing (HPC):

It has been observed that many worlds' fastest supercomputers are supported by Nvidia hardware when calculations.

- Gaming Industry:

Realistic visual environments are created by Nvidia since it is best known for its graphics technology.

- Cloud Computing and Data Centers:

Internet storage networks are powered by Nvidia hardware since it manages massive digital information.

- Scientific Research:

One of the other roles of NVIDIA is because it used in simulations, data analysis, and complex computations in various fields.

Overall, simple graphics tools were changed by Nvidia into the most necessary parts of modern computing.

D. Advantages and Limitations

Advantages

- High Performance: Great processing power is exhibited by these high-performance portable computing machines because these machines perform tasks of parallel processing like AI and Graphics. And also, the efficiency of these computers remains superior for complex calculations.
- Innovation Leadership: An intense focus is placed on research and development while NVIDIA creates CUDA and AI technologies. Because NVIDIA prioritizes discovery, it remains ahead of other competing organizations.
- Wide Applications: Artificial deep learning is utilized in many domains including gaming, AI research, automotive applications, medical imaging, and healthcare. It has been observed that AI research benefits significantly from these applications.
- Market Dominance - NVIDIA holds a large share of the market sector even though it serves a specific specialized area for GPU and AI hardware.

Limitations

- High Cost: Items manufactured by NVIDIA have prices which are higher than many other available items.
- Vendor Lock-in: technology like CUDA functions only when users operate on NVIDIA hardware. Because of this restriction, it creates dependency for the users.

- Power Consumption: Great amounts of electricity are consumed by high-performance GPUs during operation. Because these high-performance GPUs require much electricity, the operational cost increases.
- Limited Open Support - Open-source help is insufficient for the steering software components and programs for this hardware. Support is limited because the internal designs remain protected by NVIDIA.

➤ **Dell Technologies**

A. Background and History

Dell Technologies was established in 1984 by Michael Dell in the United States. Initially the company produced IBM-compatible personal computers and was considered among the first to sell directly to customers, driving down costs and increasing efficiency.

Dell grew quickly throughout the 1990s, and became the world's largest PC vendor in 2001. The firm even moved out of hardware over the years with the acquisition of Perot Systems in 2009, stepping into IT services.

B. Key Products and Technologies

- Personal Computers (PCs): Desktops and laptops for personal and business needs.
- Enterprise-class servers including PowerEdge
- Storage Devices: Storage solutions such as PowerVault and enterprise storage systems
- Networking Equipment: Network switches and such infrastructure tools
- Software & IT Services: Cloud computing, virtualization and analytics solutions
- Peripherals: Monitors, printers and others
- Dell also has a 'configure-to-order' manufacturing process that enables customers to order systems based on their configurations.

C. Roles in Modern Computing

Dell plays an important role in modern computing in the following ways:

- It manufactures personal computers that are used at home, in educational institutions, and in businesses of all sizes.
- Manufactures the IT infrastructure that is implemented by businesses to support their computing and networking needs.
- Furthermore, Dell is a major manufacturer of the computing and data center components that power the growth of cloud computing.
- It also provides the technological solutions that enable businesses to transform the way they operate and manage their resources.
- Provides IT services and support for businesses of all types around the world.

Overall, Dell Technologies has evolved from a manufacturer of computer hardware to a provider of technology solutions and services for organizations of all types.

D. Advantages and Limitations

Advantages

- Efficient direct sales and supply chain model
- Customizable systems through configure-to-order approach
- Wide range of products and services
- Strong position in enterprise and business markets
- Reliable performance
- Customizable system
- Strong customer support

Limitations

- Faced strong competition from companies like HP and Lenovo
- Profit margins negatively impacted by competition in low-cost PC market
- Product faults in the past impacted business reputation
- Dependence on PC market impacted growth during industry changes
- Some high-end modules can be expensive
- Design changes are less frequent compared to other competitors

➤ **Micron Technology**

A. Background and History

Micron was founded by engineers and local inventors in 1978 in Boise. They started as a consulting company but eventually began manufacturing their memory chips.

Throughout the company's initial years, there were many challenges. Yet, the company continued to grow by acquiring other memory chip rivals such as Elpida and they also entered into joint ventures with other giants in the semiconductor industry, such as Intel.

Nowadays, under the leadership of CEO Sanjay Mehrotra, the company has shifted its focus away from consumers (who used Crucial brand computers and gadgets) to focus on the rapidly growing areas of AI and data centers and has invested hundreds of billions of dollars into establishing new factories in the US and Singapore.

B. Key Products and Technologies

Micron manufactures two main types of memory: **DRAM** (fast and temporary) and **NAND flash** (permanent storage in SSDs and memory cards) memory. The company also manufactures solid-state drives.

It was the manufacturer of several important techniques for producing memory chips, such as atomic layer deposition and pitch double-patterning. Furthermore, it also manufactures QLC NAND (for high-density storage) and GDDR6X memory (super-fast graphics memory used in NVIDIA GPUs).

Although Micron used to sell memory drives directly to consumers under the Crucial and Ballistix brands, the company discontinued this in early 2026.

C. Role in Modern Computing

One of the three major memory makers in the world, along with Samsung and SK Hynix, is Micron. However, it is the only major company in America that manufactures both DRAM and NAND flash memory chips.

Furthermore, given recent developments with the CHIPS Act, Micron is another company that provides its memory chips to a variety of manufacturers, from data center and AI server manufacturers, to smartphone manufacturers like Apple, laptop manufacturers, graphics card manufacturers, and even car manufacturers.

Without Micron's production of memory chips, the world would have a much greater dependence upon the memory manufacturers of the Korean and Chinese countries.

D. Advantages and Limitations

Advantages:

- Access data with high speed
- Improves Multitasking
- Diverse, high-volume manufacturing across the US, Asia, and Europe.
- Strong patent portfolio and history of process innovation.
- Critical strategic asset for US technology independence.
- Deep relationships with major customers like Apple, NVIDIA, and cloud providers.

Limitations:

- Volatile memory (data is lost when power is off).
- Has limited capacity when compared to storage devices.
- Compete with companies like Samsung and SK Hynix in terms of production efficiency.
- Cyclical boom and bust industry, with Micron experiencing losses in the billions in several years.
- Legal vulnerability: Recently lost a patent case to Netlist, which cost the company \$445 million and the company was also barred from selling to Chinese clients in 2023.
- No longer owns any consumer brands, like Crucial, which was discontinued as a brand.
- Geopolitical tensions between the US and China impact the company's sales and supply chains.

➤ Toshiba

A. Background and History

Toshiba Corporation was created in 1939 with the merging of Shibaura Seisakusho and Tokyo Denki. The company, however, actually had beginnings back in 1875 and is considered a part of Japanese innovation in electrical development for the next decade.

After several decades into its life, Toshiba established itself as a leader in technological progress with such advancements in areas like consumer electronics, semiconductors, infrastructure systems.

However, the firm struggled to survive the 2010s, wrapped up in an accounting scandal and financial loss that forced a massive restructuring of the company, including sales of assets. Still, Toshiba is globally existent but its focus has evolved to the more industrial technology end products.

B. Key Products and Technologies

a) Hardware Products

- Hard disk drives (HDDs)
- Semiconductor devices
- Personal computers (historically laptops)
- Storage devices (USB drives, memory cards)

b) Infrastructure & Industrial Systems

- Power generation systems
- Elevators and escalators
- Transportation and control systems

c) Consumer Electronics

- Televisions
- Home appliances (washing machines, microwaves)
- Batteries and lighting

d) Major Technologies

- Flash Memory (NAND & NOR) – invented by Toshiba
- Early laptop computers
- DVD and HD DVD technologies
- Advanced systems like quantum cryptography and energy solutions

C. Role in Modern Computing

Toshiba plays a significant role in modern computing through:

Memory Technology: Toshiba Flash memory is found in most SSDs, smartphones and USB devices.

Storage Solutions: Hard Disk Drives (HDD) and flash storage devices are necessary computer/server data storage.

Semiconductors: It is a supplier of electronic components that Toshiba utilizes in CPU, embedded system and for other kinds of hardware.

Innovation in Computing Systems: Early laptop contributions and continued development of next-gen computing tech (e.g., quantum-based systems).

D. Advantages and Limitations

Advantages

- Strong history of innovation and technological development
- Pioneer in flash memory technology
- Wide range of products across multiple industries
- Global presence and experience in large-scale systems
- Large storage capacity
- Cost effective (HDDs)

Limitations

- Financial and management issues in recent years
- Reduced role in consumer markets (e.g., laptops and TVs)
- High competition from other technology companies
- Dependence on restructuring and business changes
- HDDs are slower than SSDs

4. INTEGRATION IN A COMPUTER SYSTEM

Modern computer architecture is based on a set of simple components, such as computer processors, memory (especially random-access memory) , storage, and computer systems and each of them performs a basic function.

The processor, which includes NVIDIA's advanced GPUs, runs applications by following instructions and doing math.

The memory (RAM), like the ones made by Micron Technology, holds active data and instructions for a short time so that the processor can get to them quickly and speed up the system.

Toshiba storage devices keep data for a long time, so files, apps, and the operating system are all safe even when the system is turned off.

Companies like Dell Technologies design complete systems that combine and manage all of these parts.

Data moves between these parts in a set order: the storage loads data into RAM, the processor processes it, and the results are either shown or saved back to storage. This constant interaction makes sure that computers work quickly and smoothly.

5. FUTURE TRENDS

Rapid increases in processing power, evolving memory architectures, and novel storage solutions are driving the future of computer hardware. These changes are designed to improve performance and efficiency, to add support for new technologies like the Internet of Things (IoT) and cloud computing.

One key trend is the relentless optimization of powerful GPUs by NVIDIA. Future generations of GPUs are anticipated to become increasingly capable of processing AI computations, DL, real-time simulations, and more energy-efficient.

In memory technology, firms like Micron Technology are progressing up the speed and efficiency curve with DDR5 and other future generation memory technologies. This will allow for a big improvement in multitasking and system reactivity.

For storage, manufacturers like Toshiba are zeroing in on high-speed solid-state drives (SSDs) and NAND flash breakthroughs. Among these advancements are increased storage speed, storage durability, storage capacity and smaller physical size.

Computer system vendors such as Dell Technologies, are meanwhile incorporating these high-end components into systems that are increasingly powerful, efficient, and ready for AI uses. More generally, the overall trend is the emergence of smarter, faster and more tightly networked computing.

6. RECOMMENDATIONS

Based on the Hardware Components mentioned in this article, a recommendation of balanced specs can be given for users based on individual computing requirements.

- For more graphic - intensive tasks such as gaming or artificial intelligence, a high - end gaming CPU or dedicated graphics card like NVIDIA is ideal.
- RAM capacity is an increasingly crucial element that's produced by several chip manufacturers, one of which is Micron.
- Toshiba is one SSD Manufacturer to choose over HDDs.
- Companies such as Dell Technologies will have to focus more on making sure that components are energy - efficient, can be upgraded for the future and meet the increasing demands of everyday consumers and professionals.

7. CONCLUSIONS

Modern computers are the combined product of multiple hardware parts such as processors, memory chips, hard drives or solid - state drives and the integrated system which they form.

Each of NVIDIA, Micron Technology, Toshiba and Dell Technologies played a vital part in these developments and helped to improve the hardware available for consumers.

Computer components are studied and seen to how the computers evolved to be faster, better and more power - efficient. Each part provides important contributions to the system performance and serve an essential role, helping computers perform all kinds of complex operations for all walks of life in areas such as education, the workplace, video gaming and AI.

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